DAD MOHAMMAD

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ACADEMIC BACKGROUND

The Islamia University of Bahawalpur — Bahawalpur, Punjab, Pakistan
Bachelor of Science in Software Engineering (6th Semester) — Expected Graduation: 2027

- **Cumulative Grade Point Average (CGPA): 3.02 / 4.00**.
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Analysis & Design, Database Engineering, Operating Systems, Machine Learning Foundations, Software Architecture, RESTful Distributed Systems.

TECHNICAL EXPERTISE

Core Languages:	Python, C++, SQL, JavaScript (ES6+), HTML5, CSS3
AI, ML & Agents:	YOLOv8, OpenCV, Microsoft Azure AI Foundry, Amazon Bedrock, Prompt Optimization, Agent Architectures
Web Frameworks:	Django, Django REST Framework (DRF), Asynchronous Request Handling
Cloud & Databases:	AWS Cloud Foundations, Microsoft Azure, IBM Cloud Concepts, SQLite, PostgreSQL, MySQL
Developer Tools:	Git, GitHub, Postman, Linux Terminal & Bash Scripting, VS Code, REST API Integration

KEY ENGINEERING & AI PROJECTS

AI-Driven Road Hazard & Pothole Detection System — Computer Vision & Web
Tech Stack: Python, YOLOv8, OpenCV, Django, SQLite, REST Architecture — GitHub Repository

- Architected an end-to-end computer vision pipeline using **YOLOv8** and **OpenCV** to detect road damage, surface distress, and potholes with millisecond latency.
- Engineered a full-stack dashboard in **Django** that maps incident coordinates, aggregates severity metrics, and feeds spatial data to an API endpoint for municipal planning.
- Optimized batch inference throughput on CPU architectures, reducing average frame processing latency by 35%.

Enterprise Document Intelligence & RAG Query System — Generative AI
Tech Stack: Python, Amazon Bedrock, Azure AI Foundry, Django REST Framework, Vector Embeddings — GitHub Repository

- Designed a modular Retrieval-Augmented Generation (RAG) platform to extract semantic context and answer domain-specific inquiries across high-volume academic and enterprise PDFs.
- Integrated foundation models through **Amazon Bedrock** and orchestrated agent workflows via **Azure AI Foundry** with guarded validation rules.
- Implemented asynchronous RESTful endpoints with **DRF** to enable real-time streaming answers and context citation tokens.

Scalable Multi-Tenant Service Portal & API Gateway — Full-Stack Backend
Tech Stack: Python, Django, Postman, Relational DB Design, RBAC — Production Prototype

- Constructed relational database models with normalized multi-tiered tables, implementing strict role-based access control (RBAC) and JWT token validation.
- Authored comprehensive API test suites using Postman and automated test scripts to ensure data consistency and 99.8% uptime reliability across user requests.

VERIFIED PROFESSIONAL CERTIFICATIONS

Specialized credentials issued by premier institutions and industry leaders via Coursera with verified cryptographic records.

Amazon Bedrock Customization & Automation
Amazon Web Services (AWS) Jul 2026
ID: 0E150AW0K2RM • Foundation Model Tuning & Agents

AI Agent Fundamentals with Azure AI Foundry
Microsoft Sep 2026
ID: 3DUGEFKNDIY3 • Autonomous Multi-Agent Systems

AWS Generative AI for Developers
Amazon Web Services (AWS) Jul 2026
ID: 4CERFDM4RE4B • GenAI Model Orchestration & Cloud API

Intro to Artificial Intelligence on Azure
Microsoft Jul 2026
ID: 4LZXZEPT4BXA • Cloud AI Cognitive Services

AI & Deep Learning Concepts & Applications
Simplilearn Jul 2026
ID: 0DBUH4N657ZZ • Neural Networks, Computer Vision & NLP

Linear Algebra for ML & Data Science
DeepLearning.AI Jul 2026
ID: 9MCJ89VWE0JG • Matrix Operations, Eigenvalues & PCA

Probability & Statistics for ML & Data Science
DeepLearning.AI Jul 2026
ID: AC67ZFW3JXYV • Hypothesis Testing & Bayes Inference

Introduction to APIs
Sage Publications Oct 2026
ID: 22BE8TVSUF31 • Endpoints, Webhook Paradigms & Microservices

Introduction to Cloud Computing
IBM Skills Network Oct 2026
ID: 85UCJXX11FIZ • Cloud Native Architecture, IaaS, PaaS, Hybrid Deployments

LEADERSHIP, ENTREPRENEURSHIP & CAMPUS INITIATIVES

Co-Founder & Lead Systems Developer 2025 – Present
PentaxSolution (Software Development & Tech Consulting) Pakistan

- Co-founded a student tech initiative delivering full-stack web applications, relational database architectures, and API integrations for departmental and client workflows.
- Direct code reviews, CI/CD repository structure on GitHub, and ensure high coding standards adhering to modular architectural patterns.

Technical Volunteer & Campus Ambassador 2024 – Present
Department of Software Engineering, Islamia University of Bahawalpur Bahawalpur

- Assisted junior semester cohorts through hands-on tutorial sessions covering Python syntax, algorithmic thinking, and Git version control workflows.
- Actively represented regional students from Balochistan in campus peer-mentorship networks, promoting technical literacy and collaborative software development.

ACADEMIC RESEARCH INTERESTS

Primary Focus Areas: Applied Artificial Intelligence, Multi-Agent LLM Orchestration, Computer Vision for Autonomous Infrastructure Assessment, and Cloud-Native Distributed Software Systems. Aiming to pursue a research-intensive Master of Science (MS/MSc) program focused on real-world AI deployment and scalable machine intelligence.

LANGUAGES

Languages: **English** (Professional / Academic Proficiency), **Urdu** (Native / Fluent), **Balochi** (Mother Tongue), **Brahvi** (Native / Fluent)